Mo Li, Ph.D.

Academic Appointment	**Assistant Professor of Statistics**	August 2023–Present
	Department of Mathematics, University of Louisiana at Lafayette	
	439 Maxim Doucet Hall	mo.li@louisiana.edu
	Lafayette, LA 70504-4330	github.com/mli171

Research Interests

Computational Statistics, Genetic Algorithms, and Changepoint Detection
Statistical Inference for Dependent and Categorical Time Series
Statistical Methods for Microbiome and TCR Sequencing Data

Education

Ph.D., Systems Modeling and Analysis (Statistics Concentration), December 2021
Virginia Commonwealth University, Richmond, VA

M.S., Operations Research, May 2018
Virginia Commonwealth University, Richmond, VA

B.S., Statistics and Biology, *summa cum laude*, May 2015
University of New Mexico, Albuquerque, NM

B.S., Biology, July 2013
Northwest University, Xi'an, China

Research Experience

Assistant Professor, Department of Mathematics — August 2023–Present
University of Louisiana at Lafayette, Lafayette, LA

Postdoctoral Fellow, Department of Biostatistics — October 2021–July 2023
Johns Hopkins University, Baltimore, MD

Papers & Publications

Peer-Reviewed Journal Publications

(† First/Co-first author; ‡ student mentee; * corresponding author)

1. Satten, G. A., **Li, M.**, Zhao, N., and Strawderman, R. L. (2026). R-estimation with right-censored data. *Lifetime Data Analysis*. Accepted for publication.
2. **Li, M.*** and Lu, Q. (2026). changepointGA: An R package for fast changepoint detection via genetic algorithms. *The R Journal*. Online.
3. Meyerson, N. S., Crifasi, C. K., **Li, M.**, Zhao, N., and Webster, D. W. (2026). Assessing differences in measurement of nonfatal firearm crime in the Uniform Crime Report Summary Reporting System and National Incident-Based Reporting System. *Journal of Urban Health*, 103(4), 639–648.
4. Peter, J., **Li, M.**, Brennan, J., Menendez, H., and Moradi Rekabdarkolaee, H. (2026). 2024 ASAS-NANP Symposium: Mathematical modeling in animal nutrition: Review of time series techniques and machine learning models for advancing livestock production. *Journal of Animal Science*, skaf439.
5. Li, R., **Li, M.**, and Zhao, N. (2025). A mixed-effect kernel machine regression model for integrative analysis of alpha diversity in microbiome studies. *Genetic Epidemiology*, **49**(1), e22596.
6. Deng, Q., Zhang, Y., **Li, M.**, Zhang, S., and Ding, Z. (2024). Efficient eigen-decomposition for low-rank symmetric matrices in graph signal processing: An incremental approach. *IEEE Transactions on Signal Processing*.

7. **Li, M.**, Hua, X., Li, S., Wu, M. C., and Zhao, N. (2024). A multi-bin rarefying method for evaluating alpha diversities in TCR sequencing data. *Bioinformatics*, **40**(7), btae431.
8. **Li, M.** and Lu, Q. (2022). Changepoint detection in autocorrelated ordinal categorical time series. *Environmetrics*, **33**(7), e2752.
9. **Li, M.**, Tyx, R. E., Rivera, A. J., Zhao, N., and Satten, G. A. (2022). What can we learn about the bias of microbiome studies from analyzing data from mock communities? *Genes*, **13**(10), 1758.
10. Wang, X., **Li, M.**, Yang, Z., Li, H., Wang, Y., Tang, W., Wu, Y., Xiao, P., Jiang, S., Shi, Q., et al. (2021). Comparison of the protective effect of different mild therapeutic hypothermia temperatures on intestinal injury after cardiopulmonary resuscitation in rats. *Shock*, **56**(3), 450–460.
11. Li, H., **Li, M.**, Yan, J., Gao, L., Zhou, L., Wang, Y., Li, Q., Wang, J., Chen, T., Wang, T., et al. (2020). Voriconazole therapeutic drug monitoring in critically ill patients improves efficacy and safety of antifungal therapy. *Basic & Clinical Pharmacology & Toxicology*, **127**(6), 495–504.

Manuscripts Under Revision

1. **Li, M.**, Lu, Q., and Wang, X. *Changepoint analysis of hourly sky-cloudiness conditions in Canada.* Revision submitted to *The Annals of Applied Statistics*.
2. Satten, G. A.†, **Li, M.**†*, and Zhao, N. * *CAFT: A compositional log-linear model for microbiome data with zero cells.* Revision submitted to *Bioinformatics*.
3. Ajadi, N., **Li, M.**, and Sang, Y. *Exponential K-sample hypothesis testing.* Revision in preparation for *Journal of Applied Statistics*.
4. Williams, M., **Li, M.**, Su, Y., and Lu, Q. *A conditional least squares parameter estimation method for the AOP model.* Revision in preparation for *Computational Statistics & Data Analysis*.

Manuscripts Under Review

1. **Li, M.** *Structured and consensus-guided island model genetic algorithm for multiple changepoint detection.* Submitted to *Computational Statistics*.
2. Siwakoti, C.‡, Moradi Rekabdarkolaee, H., Brennan, J., Menendez, H., and **Li, M.*** *Multiple changepoint detection in linear mixed-effects models via genetic algorithms with application to winter heifer growth.* Submitted to *Journal of Agricultural, Biological and Environmental Statistics*.
3. Pilla, S. J., **Li, M.**, Zhao, N., Zeger, S. L., Mathioudakis, N., Balasubramanyam, A., Knowler, W., Hampe, C., Lernmark, A., Pietropaolo, S., Clark, J., and Maruthur, N. *Islet autoantibodies in adults with type 2 diabetes predict long-term glycemic control but not clinical outcomes.* Submitted to *The Lancet Diabetes & Endocrinology*.

Selected Manuscripts in Preparation

- *Data-Adaptive Multi-Bin Rarefying for Alpha Diversity Association Analysis under Heterogeneous Sequencing Depth (MBRarefy).*
- *Pairwise Composite Conditional Least Squares Estimation for Autoregressive Moving Average Ordered Probit Models (PCCLS).*
- *Accommodating Differential Read Depth in Analyses of Virome Diversity.*

Published Conference Abstracts

1. Shore, X., **Li, M.**, Myers, O., and Sood, A. (2026). A58-14: The association between magnesium depletion score and chronic obstructive pulmonary disease: A cross-sectional analysis of the NHANES database 2017–2020. *American Journal of Respiratory and Critical Care Medicine*,

212(Suppl. 1).

2. Maruthur, N. M., Knowler, W. C., Balasubramanyam, A., Hampe, C. S., Pietropaolo, S. L., Anderson, A. M., **Li, M.**, Zhao, N., Zeger, S. L., Ramelius, A., Bennet, R., Lernmark, Å., Mathioudakis, N. N., Clark, J. M., and Pilla, S. J. (2023). Clinical features associated with glutamic acid decarboxylase autoantibodies (GADA) in adults with type 2 diabetes. *Diabetologia*, **66**(Suppl. 1), S227.
3. Pilla, S. J., Knowler, W. C., Balasubramanyam, A., Hampe, C. S., Pietropaolo, S. L., Anderson, A. M., **Li, M.**, Zhao, N., Zeger, S. L., Ramelius, A., Bennet, R., Lernmark, Å., Mathioudakis, N. N., Clark, J. M., and Maruthur, N. M. (2023). Glutamic acid decarboxylase autoantibodies (GADA) in type 2 diabetes predict long-term glycaemic control but not complications. *Diabetologia*, **66**(Suppl. 1), S227–S228.
4. Pilla, S. J., Knowler, W. C., Balasubramanyam, A., Hampe, C. S., Pietropaolo, S. L., Anderson, A. M., **Li, M.**, Zhao, N., Zeger, S. L., Ramelius, A., Bennet, R., Lernmark, Å., Mathioudakis, N. N., Clark, J. M., and Maruthur, N. M. (2023). Clinical features associated with glutamic acid decarboxylase autoantibodies (GADA) in adults with type 2 diabetes. *Diabetes*, **72**(Suppl. 1), 1388-P. doi:10.2337/db23-1388-P.
5. Pilla, S. J., Knowler, W. C., Balasubramanyam, A., Hampe, C. S., Pietropaolo, S. L., Anderson, A. M., **Li, M.**, Zhao, N., Zeger, S. L., Ramelius, A., Bennet, R., Lernmark, Å., Mathioudakis, N. N., Clark, J. M., and Maruthur, N. M. (2023). Glutamic acid decarboxylase autoantibodies (GADA) in type 2 diabetes do not predict long-term diabetes complications. *Diabetes*, **72**(Suppl. 1), 1308-P. doi:10.2337/db23-1308-P.
6. Pilla, S. J., Knowler, W. C., Balasubramanyam, A., Hampe, C. S., Pietropaolo, S. L., Anderson, A. M., **Li, M.**, Zhao, N., Zeger, S. L., Ramelius, A., Bennet, R., Lernmark, Å., Mathioudakis, N. N., Clark, J. M., and Maruthur, N. M. (2023). Glutamic acid decarboxylase autoantibodies (GADA) in type 2 diabetes predict glycemic control over 16 years. *Diabetes*, **72**(Suppl. 1), 1387-P. doi:10.2337/db23-1387-P.

Statistical Software

- `changepointGA` — R package for genetic-algorithm changepoint detection; CRAN and GitHub.
- `GAReg` — R package for discrete optimization in regression; CRAN and GitHub.
  Selected among the February 2026 "Top 40 New CRAN Packages" (40 of 255).
- `CAFT` — R package for differential abundance analysis in microbiome data; CRAN and GitHub.
- `RAFT` — R package for rank-based accelerated failure-time inference; GitHub.
- `MBRarefy` — R package for multi-bin rarefaction; GitHub.
- `TSAOP` — R package for ordered probit model for ordinal time series; GitHub.

Sponsored Research

Funded Research Grants

T Cell Receptor Diversity Analysis and Clonotype Assessment in Immunological Research 2025–2028

Principal Investigator (single PI). Louisiana Board of Regents Support Fund, LEQSF(2025–28)-RD-A-18. Total project amount: $192,295 ($143,370 external support).

Stability-Guided Genetic Algorithm Changepoint Detection for HRV-Based Stress Monitoring 2026–2027

Principal Investigator (single PI). Vernon and Ruby Langlinais Research Fund, Ray P. Authement College of Sciences, University of Louisiana at Lafayette. $10,000.

Pending and Recently Submitted Proposals

Data-Efficient Inference under Heterogeneous Sequencing Depth in Biotechnology October 2026

Single-PI **internal preproposal** for NSF EPSCoR Emerging Research Fellows; submitted to University of Louisiana at Lafayette institutional competition.

Honors & Awards

Louisiana Impact Sustainable Development Research Award, $2,000, University of Louisiana at Lafayette	2025
The 10th Workshop on Biostatistics and Bioinformatics Travel Grant, Georgia State University	2025
Faculty Professional Development Travel Grant, $1,000, University of Louisiana at Lafayette	2025
SRCOS Summer Research Conference Travel Award, $600, University of Louisiana at Lafayette	2024
Faculty Research Travel Grant, $1,500, University of Louisiana at Lafayette	2024
Mu Sigma Rho, National Statistical Honor Society	2016

Teaching Experience

University of Louisiana at Lafayette — Instructor of Record

Course, semester, and student evaluation (5-point scale, where available):

STAT 521, Regression Analysis	Fall 2026	In progress
STAT 520, Probability Theory	Fall 2026	In progress
STAT 325, Introduction to Probability and Statistics	Summer 2026	Pending
STAT 522, Experimental Design	Spring 2026	4.80/5.00
STAT 528, Applied Time Series	Fall 2025	5.00/5.00
STAT 521, Regression Analysis	Fall 2025	4.60/5.00
STAT 522, Experimental Design	Spring 2025	4.90/5.00
STAT 521, Regression Analysis	Fall 2024	4.60/5.00
STAT 417, Biometry	Fall 2024	4.00/5.00
STAT 417, Biometry	Spring 2024	4.40/5.00
STAT 417, Biometry (two sections)	Fall 2023	4.60/5.00

Virginia Commonwealth University — Instructor of Record

MATH 141, Algebra with Applications	Spring 2021	4.33/5.00
MATH 141, Algebra with Applications	Fall 2020	4.28/5.00
MATH 131, Introduction to Contemporary Mathematics	Spring 2019	4.69/5.00
MATH 151, Pre-Calculus	Summer 2018	4.51/5.00

Virginia Commonwealth University — Teaching Assistant

- STAT 314, Applications of Statistics: Spring 2020, Spring 2018, Fall 2016, Spring 2016, Fall 2015.
- MATH 151, Pre-Calculus: Fall 2018, Fall 2017.
- STAT 514, Mathematical Statistics II: Spring 2017.

Graduate Mentoring

Primary Research Mentoring

- Imanthi Jayathilake — graduate research in time-series downsampling and autoregressive ordered probit models.

- Chhabi Siwakoti — graduate research in multiple changepoint detection for linear mixed-effects models.

Ph.D. Dissertation Committees

Ibrahim Adenekan — Member	Summer 2026
Justin Andrew Dunnam — Member	Spring 2026
Gboyega Adepoju — Member	Spring 2025
Sameera Hewage — Member	Spring 2025
Md Monzur Murshed — Member	Spring 2024
Md. Mahadi Hasan — Member	Spring 2024

Master's Thesis Committees

Samuel Simon — Member	Fall 2025
Imanthi Jayathilake — Member	Fall 2025
Madushanka Kankanamge — Member	Fall 2025
Oluwafemi Afolabi Jegede — Member	Fall 2025
Benjamin Onyemaechi Omele — Member	Fall 2025
Sashen D. Fernando — Chair	Spring 2025
Justin Andrew Dunnam — Member	Fall 2024

Talks & Presentations

Invited Talks and Seminars

1. "Multiple Changepoint Detection in Linear Mixed-Effects Models via Genetic Algorithms with Application to Winter Heifer Growth." 20th International Joint Conference CFE-CMStatistics, HTW Berlin (Berlin University of Applied Sciences), Berlin, Germany. December 12–14, 2026. *Invited talk (scheduled).*
2. "CAFT: A Compositional Log-Linear Model for Microbiome Data with Zero Cells." International Conference on Advances in Interdisciplinary Statistics and Combinatorics, University of Louisiana at Lafayette, Lafayette, LA. October 30–November 1, 2026. *Invited talk (scheduled).*
3. "Genetic Algorithms in Regression." 9th International Conference on Econometrics and Statistics (EcoSta 2026), Ryukoku University, Kyoto, Japan. August 8–10, 2026. *Invited talk.*
4. "Changepoint Detection in Categorical Time Series with Application to Daily Total Cloud Cover in Canada." 19th International Joint Conference CFE-CMStatistics, Birkbeck, University of London, UK. December 13–15, 2025. *Invited talk.*
5. "Changepoint Detection in Autocorrelated Ordinal Categorical Time Series." AMS Graduate Student Chapter, University of Louisiana at Lafayette. November 18, 2025. *Invited talk.*
6. "A Compositional Log-Linear Model for Microbiome Data with Zero Cells." School of Mathematical and Statistical Sciences, University of Texas Rio Grande Valley. November 7, 2025. *Invited talk.*
7. "Genetic Algorithm Application in Changepoint Detection." Department of Mathematics, University of Louisiana at Lafayette. April 11, 2025. *Seminar.*
8. "Rank-Based Compositional Analysis Using Log-Linear Models for Microbiome Data with Zero Cells." Department of Mathematics, University of Louisiana at Lafayette. April 3, 2025. *Colloquium.*
9. "A Multi-Bin Rarefying Method for Evaluating Alpha Diversities in TCR Sequencing Data." 18th International Joint Conference CFE-CMStatistics, King's College London, UK. December 14–16, 2024. *Invited talk.*

10. "A Multi-Bins Rarefying Method for Testing the Alpha Diversities in TCR Sequencing Data." Department of Mathematics and Statistics, South Dakota State University. November 22, 2024. *Invited talk.*
11. "Changepoint Analysis of Hourly Sky-Cloudiness Conditions in Canada." Department of Mathematics, University of Louisiana at Lafayette. November 17, 2023. *Seminar.*
12. "Statistical Methods for Discrete Data with Applications in Climate and Microbiome Studies." Department of Mathematics and Statistics, Utah State University. February 14, 2023. *Invited talk.*
13. "Statistical Methods for Discrete Data with Applications in Climate and Microbiome Studies." Department of Nutrition and Health Science, Ball State University. February 6, 2023. *Invited talk.*
14. "Statistical Methods for Discrete Data with Applications in Climate and Microbiome Studies." Department of Mathematics, University of Louisiana at Lafayette. January 17, 2023. *Invited talk.*
15. "Changepoint Detection in Multivariate Time Series via Deep Learning." Department of Statistical Sciences and Operations Research, Virginia Commonwealth University. October 28, 2018. *Seminar.*

Contributed Conference Talks

1. "CAFT: A Compositional Log-Linear Model for Microbiome Data with Zero Cells." Louisiana Chapter of the American Statistical Association Spring Meeting, Baton Rouge, LA. April 10, 2026. *Contributed talk.*
2. "A Multi-Bin Rarefaction Method for Evaluating Alpha Diversities in TCR Sequencing Data." Louisiana Chapter of the American Statistical Association Spring Meeting, New Orleans, LA. May 19, 2025. *Contributed talk.*
3. "A Multi-Bins Rarefaction Method for Testing the Alpha Diversities in TCR Sequencing Data." 5th National Big Data Health Science Conference, Columbia, SC. February 2–3, 2024. *Conference presentation.*

Conference Posters

1. "Rank-Based Compositional Analysis Using Log-Linear Models for Microbiome Data with Zero Cells." 10th Workshop on Biostatistics and Bioinformatics, Georgia State University, Atlanta, GA. May 10, 2025. *Poster.*
2. "Changepoint Analysis of Daily Sky-Cloudiness Conditions in Canada." SRCOS Summer Research Conference, Clemson University, Clemson, SC. June 3, 2024. *Poster.*

Invited Presentation (Not Delivered)

"CAFT: A Compositional Log-Linear Model for Microbiome Data with Zero Cells." International Chinese Statistical Association (ICSA) Applied Statistics Symposium, George Mason University, Arlington, VA, June 14–17, 2026. Invited speaker and invited session chair; unable to attend owing to institutional travel-funding restrictions.

Professional Service

Conference and Professional Service

- Session chair (scheduled), "Statistical and Machine Learning Methods for Nonstandard Data Structures," International Conference on Advances in Interdisciplinary Statistics and Combinatorics, University of Louisiana at Lafayette, October 30–November 1, 2026.

- Conference organizer, International Conference on Advances in Interdisciplinary Statistics and Combinatorics, University of Louisiana at Lafayette, October 30–November 1, 2026.
- Organizing committee member, Lloyd Roeling Conference: Statistics 2025, October 24–25, 2025.
- Special issue topic coordinator, *Frontiers in Genetics*, "TCRpred: Incorporating T-Cell Receptor Repertoire for Clinical Outcome Prediction," November 21, 2024.
- Reviewer, Best Paper Award Competition, 2024 INFORMS Conference on Quality, Statistics, and Reliability (ICQSR), June 15, 2024 (four papers).
- Judge, Student Distinguished Paper Award Competition, ASA Section on Statistics in Genomics and Genetics (SSGG), January 14, 2024 (five papers).

Journal Refereeing

Computational Statistics	2026
BMC Medical Research Methodology	2023, 2026
Journal of Big Data	2026
Agronomy Journal	2026
Communications in Statistics: Case Studies, Data Analysis and Applications	2025
PLOS ONE	2024, 2025
Journal of Statistical Theory and Practice	2025
Frontiers in Genetics	2024

UNIVERSITY SERVICE

Data Science Committee, Member	2023–2026
Statistics Seminar Committee, Member	2025–2026
Comprehensive Examination Committee, Applied Statistics — Member	2026
Comprehensive Examination Committee, Experimental Statistics — Chair	2025–2026
Comprehensive Examination Committee, Experimental Statistics — Member	2023–2024
Ad hoc proposal review committee, LSU A&M B.S.S. in Applied Data Science Program	April 2025
Graduate Program Academics and Advising Appeals Committee, Member	2024–2026
Undergraduate Program and Curriculum Committee, Member	2023–2026